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Toward Complementary Intelligence: Integrating Cognitive and Machine AI

Cleotilde Gonzalez and Tailia Malloy
Department of Social and Decision Sciences, Carnegie Mellon University

Abstract

This article calls for complementary human-AI intelligence. Rather than redefining intelligence to fit machine capabilities, we argue for designing AI that complements and extends human cognition. We distinguish between *cognitive AI*, which is grounded in cognitive science to model human perception, learning, and decision-making, and *machine AI*, which achieves large-scale performance through data-driven optimization. Building on advances in machine learning alignment and human-AI complementarity, we propose an integrative framework that connects cognitive and machine AI across four routes: *embedding integration*, aligning human and machine representations; *instruction encoding*, using machine AI to translate goals into cognitive AI; *training agents*, using cognitive AI to guide and train machine AI through human-like data; and *coevolving agents*, enabling cognitive and machine AI to coadapt and improve together over time. These integration routes provide a foundation for *complementary intelligence*: systems that combine human interpretability with machine scalability and precision to enhance trust, adaptability, and human agency in complex sociotechnical environments.

Keywords

cognitive-machine integration, human-AI complementarity, cognitive alignment, intelligence, human-AI alignment

Definitions of intelligence have evolved with technology. Classic accounts emphasized general ability and adaptability (e.g., Neisser et al., 1996; Spearman, 1961). Successive breakthroughs, from early engines to contemporary large language models, shifted benchmarks toward machine capabilities, often narrowing what is treated as uniquely human. Instead, we frame intelligence as *complementarity*: joint human-machine reasoning, learning, and adaptation (Vaccaro et al., 2024).

In an effort to achieve complementary human-machine intelligence, recent work has distinguished between two forms of AI: *cognitive AI* and *machine AI* (Gonzalez & Heidari, 2025). Cognitive AI draws from psychology, cognitive science, and neuroscience to model the mechanisms by which humans perceive, learn, and decide. This is achieved through the application of cognitive architectures and models that aim to replicate how people think and make decisions relying on cognitive theories. In contrast, machine AI arises from computer science and machine learning, focusing on extracting statistical regularities from large data sets to optimize performance, typically without explicit models of human cognition. As Gonzalez and Heidari (2025) argued, the path forward lies not in choosing between these paradigms but in *integrating* them—developing systems in which cognitive AI and machine AI inform, constrain, and enhance one another to support and empower human decision makers.

This integration is timely because both cognitive science and machine learning are converging on a shared challenge: alignment between humans and machines. Across domains from neuroscience to deep learning, researchers now investigate whether artificial systems form internal representations that are structurally or conceptually aligned with human cognition (Rane et al., 2024; Sucholutsky et al., 2023; Wynn et al., 2024). Representational alignment refers to the degree to which humans and machines process complex information in the formation of constrained internal representations and

Corresponding Author:
Cleotilde Gonzalez, Department of Social and Decision Sciences, Carnegie Mellon University
Email: coty@cmu.edu

measure the similarity of those representations (Mahner et al., 2025). This type of alignment is distinct from conceptual alignment, which extends this effort to human-meaningful abstractions (Rane et al., 2024) as well as behavioral alignment, which focuses on improving social and ethical outcomes of the use of AI (Ji et al., 2023).

Building on this convergence, this article advances the idea that cognitive AI and machine AI can be integrated to create hybrid systems that are not only intelligent but also aligned, interpretable, and complementary to human cognition. We outline four concrete integration routes between cognitive AI and machine AI that aim to move beyond the question of whether machines can match human intelligence toward designing systems in which cognitive AI and machine AI integrate to enhance human-machine collaborations.

Cognitive and Machine AI

The pursuit of AI has evolved along two complementary traditions: machine AI, which achieves scalable performance through data-driven optimization, and cognitive AI, which models the mechanisms of human thought and behavior. Understanding how these approaches differ and how they can be integrated is essential for achieving a new level of intelligence that is complementary to human intelligence.

Machine AI: pattern learning at scale

Machine AI systems are fundamentally data-driven. They optimize prediction, recognition, and control through large-scale computation and have become the foundation of modern assistive technologies such as recommendation systems, classification tools, and generative models (Park et al., 2023). By leveraging deep learning, reinforcement learning, and probabilistic inference, machine AI extracts statistical regularities from vast data sets to achieve scalability, precision, and adaptability in closed, well-defined tasks. These systems process immense information spaces, optimize for accuracy, and maintain consistency without fatigue or drift.

However, this power comes with limitations. Many models remain opaque ("black boxes"), lack causal understanding, and degrade under distributional shifts. Posttraining alignment methods such as reinforcement learning with human feedback fine-tune existing models using large data sets of human preferences for model output without establishing a shared internal structure with human cognition (Ouyang et al., 2022). As a result, machine AI is weakly related to how humans process information, form representations, and use their experience when learning and making decisions (Gonzalez & Heidari, 2025; Rane et al., 2024).

This weak relationship has motivated a rapidly growing body of work on representational alignment that focuses on methods for increasing the alignment of human and machine representations and reasoning structure, such as through cotraining human and machine representations (Sucholutsky et al., 2023; Wang et al., 2024; Wynn et al., 2024). These advances highlight that AI systems must align with humans not only in observable behavior but also in their underlying representations and reasoning processes to truly complement human intelligence.

Although machine AI achieves extraordinary performance through scale and optimization, it often does so without understanding or transparency. Cognitive AI provides a complementary foundation by grounding machine AI in human reasoning. Whereas machine AI focuses on learning from data, cognitive AI models the mechanisms by which humans learn from experience.

Cognitive AI: modeling and aligning with human cognition

Cognitive AI builds models of perception, memory, learning, and decision-making grounded in unified theories and architectures (Anderson et al., 1997; Newell, 1994). By encoding explicit, interpretable representations and constraints, these models align naturally with human reasoning and support collaborative, transparent interaction (Gonzalez, 2024).

Unlike data-driven systems that optimize for performance without interpretability, cognitive AI embodies explicit, theory-based representations that trace how knowledge is encoded, retrieved, and applied across contexts. This design enables psychological validity and natural human alignment, reflecting the same constraints, biases, and adaptive strategies that guide human cognition. In doing so, cognitive AI offers a foundation for systems that collaborate with humans through shared models of understanding rather than opaque statistical associations.

Cognitive AI has several notable strengths: It can learn from limited experience, generalize across related tasks, reason about causality, and provide interpretable mechanisms for decision-making (Dietz et al., 2022; Lake et al., 2017). However, it faces important challenges. Cognitive models are computationally constrained, often rely on handcrafted or simplified representations, and scale poorly to high-dimensional, real-world data. In contrast to machine AI, which optimizes performance, cognitive AI prioritizes explanatory fidelity, capturing how humans think, including the replication of human limitations. These trade-offs have limited its predictive power and scalability relative to large-scale machine AI systems.

Table 1. Cognitive and Machine AI Strengths and Weaknesses

	Cognitive AI	Machine AI
Core focus	Models human cognition—perception, memory, learning, and decision-making—using psychologically grounded theories	Optimizes data-driven performance through statistical learning, deep neural networks, and large-scale computation
Primary strengths	• Interpretable and theory-based (grounded in human cognitive architectures) • Captures mechanisms of human reasoning, biases, and learning dynamics • Enables explainable and human-aligned decision-making • Supports transfer and generalization in dynamic, uncertain, or data-sparse environments	• Scales to massive data and complex, high-dimensional tasks • Excels at pattern recognition, prediction, and optimization • Achieves superhuman accuracy in constrained domains (e.g., image recognition, language modeling) • Continuously improves via self-supervised and reinforcement learning • Highly adaptable to new architectures and modalities
Primary weaknesses	• Limited scalability for high-dimensional, data-rich problems • Slower generalization to novel tasks without remodeling • Psychological fidelity prioritized over raw performance • May lack robustness when interacting with opaque machine learning systems	• Opaque and difficult to interpret ("black box") • Lacks grounding in human cognition or decision processes • Brittle under distributional shift; lacks causal understanding • Struggles with transfer and reasoning outside of training data

Cognitive and machine AI: strengths and weaknesses

The strengths and limitations of cognitive and machine AI reveal distinct yet complementary pathways toward a new level of AI aligned with human cognition (see Table 1). Their integration promises systems that combine human-like understanding with machine-level precision.

In what follows, we discuss how cognitive and machine AI can be integrated to reach a new level of intelligence that is aligned, interpretable, and complementary to human cognition. We outline four integration routes: *embedding integration*, in which human and machine representations are aligned; *instruction encoding*, in which machine AI is used to encode instructions for cognitive AI, translating naturalistic inputs into structured cognitive representations; *training agents*, in which cognitive models provide psychologically grounded priors or data to train machine AI; and *coevolving agents*, in which cognitive and machine AI systems dynamically exchange information and goals to support adaptive human decision-making.

Integration of Cognitive and Machine AI

Bridging cognitive and machine AI promises systems that not only perform but also explain, adapt, and align the core of what we call complementary intelligence. Integrating cognitive and machine AI offers a pathway toward complementary intelligence, a synthesis in which cognitive AI provides human-centered understanding and goal alignment while machine AI contributes computational scalability and representational power (Fig. 1). Cognitive AI can guide the training and fine-tuning of machine AI models toward more human-aligned representations and causal reasoning. Conversely, machine AI can enhance cognitive AI's applicability by forming representations of complex information or by imposing an inductive bias on model performance by using instructions.

Such integration moves beyond the pursuit of AI as a replication of human cognition or an isolated optimization engine. Instead, it envisions a hybrid form of intelligence—one that learns with humans, reasons like humans, and scales beyond human limitations. This complementary intelligence is a foundation for cooperative, trustworthy, and human-centered AI systems capable of shared reasoning and adaptive decision-making in dynamic environments (Gonzalez & Heidari, 2025).

Integrating cognitive and machine AI requires more than combining models or data; it involves linking *representations*, *processes*, and *purposes* so that human-biased and constrained reasoning can coexist with machine-level scalability and adaptability. Table 2 summarizes examples of current research across these categories. Each route represents a distinct mechanism for aligning the representational, learning, or decision-making spaces with both cognitive AI and machine AI.

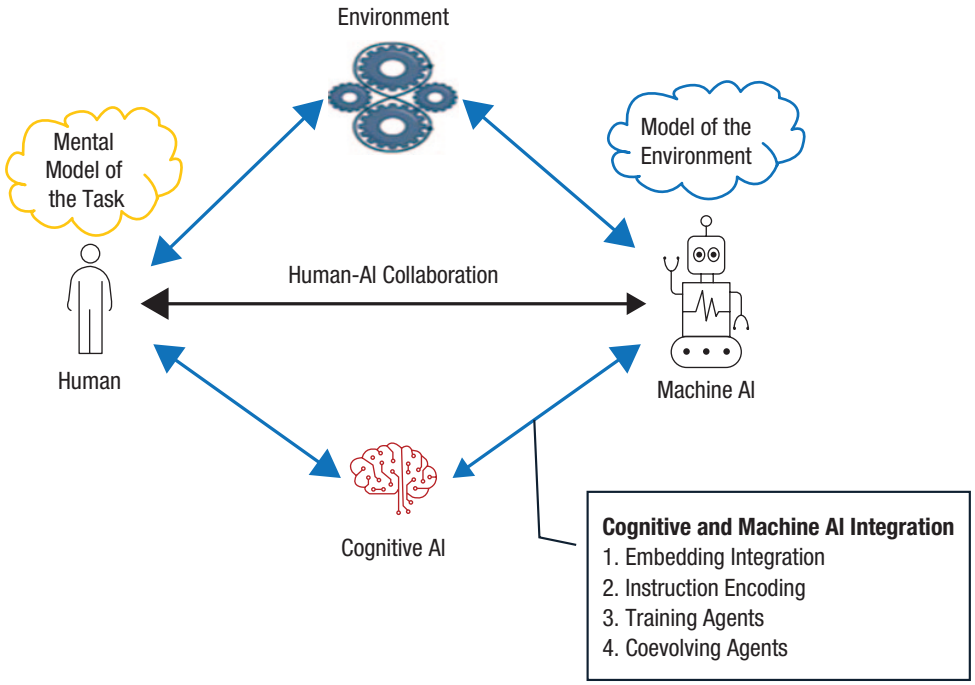


Fig. 1. Cognitive and machine AI integration for human-AI complementarity.

Whereas some categories have been explored extensively in recent research, others remain relatively underexamined. We draw on examples from our own work and related studies to illustrate these ideas. All four categories represent valuable directions for future inquiry, with significant potential for cross-category insights and integration, although additional categories may also emerge as promising avenues of research.

Embedding integration

Traditional cognitive modeling requires the creation of handcrafted representations of stimuli, which are sometimes derived from verbal protocols or by observation of experts performing a task. Deciding what attributes to represent in a cognitive model takes considerable effort and is a significant degree of freedom in the process of building a cognitive representation of a task. Additionally, cognitive models often do not account for "instructions" provided in a task and rely on the prepopulation of memory or other assumptions to account for initial information often generated by instructions (Nguyen et al., 2023).

Modern foundation models learn dense, task-general representations that capture the ways in which humans leverage language in learning and decision-making. These representations are referred to as embeddings that take the form of a list of floating-point numbers that correspond to a high-dimensional vector. When these embeddings are used as inputs to cognitive models such as instance-based learning (Gonzalez et al., 2003) or Bayesian learning (Jacobs & Kruschke, 2011), they provide rich, reusable features while the cognitive layer contributes mechanisms, interpretability, and alignment with human reasoning. This route aligns what the model uses in reasoning (its representational structure) with how humans reason, improving both fit to behavior and explanatory power. Empirically, representational alignment is measured by comparing model and human learning and decision-making through observation as well as by theoretically evaluating representation structures and concept spaces (Roads & Love, 2021).

Recent work shows that LLMs can be adapted into cognitive models that predict human behavior across decision tasks, especially after fine-tuning on behavioral data sets, sometimes outperforming traditional models (Malloy et al., 2024). This positions LLM embeddings as strong representational front ends for mechanism-rich cognitive models (e.g., adding memory retrieval or similarity measurement; Binz & Schulz, 2023).

Beyond task performance, LLM embeddings approximate human semantic structure: They predict human similarity judgments, capture relational regularities, and can be pruned or projected to better match human-interpretable dimensions (Malloy et al., 2025). These

Table 2. Categories and Examples of Cognitive–Machine AI Integration

Category of integration	AI model type	Technique or use case	Examples
Embedding integration	Large language embedding models	Antiphishing training: Use LLM embeddings to represent message content and context; a cognitive decision layer (e.g., instance-based similarity or drift-diffusion with bias parameters) models user susceptibility and training effects.	Malloy et al. (2024); Malloy, Fang, & Gonzalez (2025); Malloy, Ferreira, et al. (2025); Malloy & Gonzalez (2024)
		Human categorization of language: Start with LLM embeddings; apply a cognitively grounded categorization mechanism (e.g., exemplar or rational rules) to match human cluster structure and error patterns, potentially after embedding pruning to human-interpretable subspaces.	Manrique et al. (2023); Xu & Rajivan (2024)
		Predicting learning and decisions: Combine LLM embeddings with cognitive learning dynamics (e.g., memory decay, reinforcement, aspiration thresholds) to capture trial-level choice.	Binz & Schulz (2023); Malloy et al. (2023)
	Visual generative models	Predicting human learning and decisions: Use embeddings to represent stimuli; place a cognitive learner (e.g., instance-based or Bayesian generalization) on top to reproduce human acquisition curves and generalization gradients.	Malloy & Sims (2025); Malloy & Gonzalez (2024); Wang et al. (2024)
		Representation formation and decisions: Model representation formation as embedding reshaping under task constraints (e.g., attention-weighted projections) and then pass to a decision mechanism to exhibit human-like conceptual clustering.	Du et al. (2025); Malloy & Sims (2025)
		Transfer of learning: Treat latents as shared representational priors; cognitive parameters capture task-specific control, enabling out-of-distribution generalization and transfer across related tasks.	Kondapaneni et al. (2024); Malloy et al. (2023)
Machine AI to encode instructions in cognitive AI	Agentic LLMs	Applying LLMs to guiding policies in cognitive models: Demonstrate how LLMs can serve as instructional intermediaries that translate naturalistic task descriptions into structured policies compatible with cognitive architectures.	McDonald et al. (2023); Honovich et al. (2022)
		Encoding spatial and relational instructions with LLMs: Translate phrases into vector embeddings that reflect spatial configuration and relational meaning.	Martorell (2025)
Cognitive AI to train machine AI	Synthetic data and constraints	Simulate human decisions at scale: Use cognitive models to generate human-like actions and outputs.	Aggarwal et al. (2022); Trafton et al. (2020)
		Cognitive constraints as priors: Use cognitive models to inform loss functions, priors, and reward shaping.	Murthy et al. (2025); Sense et al. (2022)
	Agentic LLMs	Agentic AI: Surveys and frameworks on instruction—or preference—tuning, and agent training contextualize how cognitive signals can replace or augment human feedback.	Liu et al. (2024); Ouyang et al. (2022); Plaat et al. (2025); Wang et al. (2024)
	Cognitive multiagent simulators	Multiagent interaction: Use cognitively scaffolded generative agents to produce dialogue/coordination trajectories and couple with cognitive diagnostics to keep policies within human-plausible bounds.	Park et al. (2023)
	Visual generative models	Human-like reinforcement learning for visual information: Use visual scenes labeled with cognitive learning models to teach perception-to-policy stacks human-consistent exploration, credit assignment, and transfer.	de Melo et al. (2022); Malloy & Sims (2025)
Coevolving agents	LLMs + cognitive tutors	Training to identify phishing emails: A cognitive model of user susceptibility (learning history, attention to cues, risk thresholds) predicts near-future errors and distills them into prompt scaffolds for an LLM tutor that generates tailored exemplars, explanations, and spaced-repetition schedules; the LLM's responses and the learner's performance flow back to update the cognitive state.	Malloy, Fang, & Gonzalez (2025)
	Multicognitive AI agent interaction	Large groups in simulated worlds: Populate a sandbox (e.g., town of agents) with cognitive AI agents (explicit memory/attention/utility mechanisms) and LLM agents (rich language/planning); the cognitive agents monitor group-level constructs (norms, diffusion of information, fatigue) and emit constraints or curricula, the LLM agents adapt plans/dialogue, and logs feed back to update cognitive states.	Park et al. (2023)
	Beyond language models	Timing and performance signals: The cognitive model schedules when and what to present while the machine learning side adapts parameters or boosting policies; online performance loops back to refine the cognitive schedule.	Shivaswamy & Joachims (2015)

Note: LLM = large language model.

studies use representation selection/reshaping before feeding features into cognitive architectures for categorization, analogy, or judgment tasks (Manrique et al., 2023). These links also appear at neural and behavioral levels: Transformer language models explain substantial variance in neural responses during language processing and correlate with behavioral measures (e.g., reading times), strengthening the case that LLM embeddings encode human-relevant predictive structure that cognitive mechanisms can leverage (Schrimpf et al., 2021).

For vision, diffusion models offer multimodal or generative embeddings that align with human judgments and neural data. Features predict human similarity measurement, suggesting they encode concept-level information that can feed cognitive mechanisms for categorization, memory, and decision (Roads & Love, 2021). Embedding such latents into cognitive models enables accounting for learning curves, attention shifts, and transfer, as shown in recent work (Kondapaneni et al., 2024; Malloy & Gonzalez, 2024).

Instruction encoding

One underexplored but potentially powerful integration route uses AI systems to translate naturalistic inputs (especially instructions, task definitions, or experiential descriptions) into structured representations that feed into cognitive models. Cognitive modeling has long grappled with the challenge of how participants use prior real-world experience or explicit instructions before beginning a learning or decision-making task. By using a machine AI front end (e.g., an LLM) to parse, encode, and structure instructional or experiential material, the cognitive layer can begin more closely aligned with the human participant's frame of reference.

For example, recent work shows that LLMs can infer underlying task definitions from a few demonstrations and generate natural language instructions that reflect that structure (Honovich et al., 2022). In decision modeling, work highlights how generative AI (including LLMs) can feed cognitive models by generating synthetic experience based on natural language instruction and thus supporting instance-based models and memory retrieval in cognitive architectures (Malloy & Gonzalez, 2024). In spatial or navigation-type tasks, research demonstrates how LLMs convert textual instructions or descriptions of spatial relationships into latent representations that correspond to navigation behavior and internal activations in grid-world tasks (Martorell, 2025). This directly illustrates how natural language instruction can be mapped into structured, model-usable forms.

Thus, a cognitive model might begin with a preparatory phase in which an LLM consumes the human-task instructions (and perhaps prior experiential text) and outputs a structured representation (feature set, memory instances, or relational graph). The cognitive architecture then uses its mechanisms (e.g., memory retrieval, similarity matching, drift-diffusion decision processes) on that structured representation to simulate human behavior (e.g., using LLM-derived representations of behavior as the initial experience of cognitive models in decision-making experiments; McDonald et al., 2023).

Training agents

A third integration path uses cognitive models to inform machine AI: supplying synthetic data, priors, and reward signals and constraining learning toward human-like behavior and reasoning. Because training machine AI typically requires vast data sets, cognitive models offer a scalable alternative: simulating human decisions in learning and decision-making tasks to generate realistic labeled data for fine-tuning or behavior cloning (Aggarwal et al., 2022; Bugbee & Gonzalez, 2022; Trafton et al., 2020). Recent surveys have positioned such synthetic, cognitively grounded data as a first-class resource for modern machine learning, particularly when privacy, bias, or data scarcity constrain real-world data collection (Liu et al., 2024).

One of the key advantages of cognitive AI is that its architecture imposes constraints grounded in theories of human cognition, limiting the behaviors the models can exhibit. For example, this constraint may form an inductive bias when generating data, resulting in a distribution of data that matches patterns observed in human behavior. These constraints help ensure the generated decisions result in large, human-like data sets for fine-tuning AI models (Sense et al., 2022). Beyond data, cognitive theories can inform loss functions and reward shaping. For example, value trade-off models (e.g., politeness as balancing goals) have been used to probe how posttraining choices change LLM trade-offs, suggesting that explicit cognitive objectives can be injected as training targets or reward components instead of only human preference labels. In the field of deep reinforcement learning, researchers are increasingly arguing for cognitive-inspired objectives and diagnostics to make agents more interpretable and human-compatible (Murthy et al., 2025).

Generative agents and related agentic LLM environments also provide rich, human-like interactive corpora. When these agents are cognitively scaffolded (memory systems, reflection, planning), their interactions yield trajectories with believable social dynamics that can bootstrap tool-use policies, dialogue strategies, and long-horizon planning for downstream models (Park et al., 2023). In visually grounded decision-making, these cognitively "shaped" corpora can fine-tune

perception-to-policy stacks to behave more like humans (Malloy & Sims, 2025). Reviews of synthetic data-driven training emphasize exactly this simulation and constraint pattern for next-generation deep learning models (de Melo et al., 2022).

Coevolving agents

Unlike one-way pipelines, coevolving systems couple cognitive and machine AI in continuous feedback loops (see Fig. 1): Each adapts to the other and to the human, updating representations, behaviors, and inputs to machine AI models over time. This framework builds on human-AI teaming and interactive learning research emphasizing mutual adaptation, shared mental models, and ongoing preference inference rather than one-shot fine-tuning. Coactive learning formalizes this process, with humans and their cognitive proxies providing incremental corrections that guide machine AI toward personalized, effective collaboration, including recent extensions to LLM personalization (Tucker et al., 2024).

Value alignment and collaboration are inherently coadaptive: The machine must infer goals, teach and be taught, and adapt as humans (and cognitive models) change their behavior. Cooperative inverse reinforcement learning makes this explicit, defining optimal policies that include active teaching and communication (Hadfield-Menell et al., 2016). Cognitive models also play a pedagogical role within these loops, structuring feedback and curricula to accelerate convergence and trust.

Emerging infrastructures for coadaptive cognitive-machine teaming (e.g., in which cognitive models track human states and feed constraints or prompts to LLM agents that, in turn, update plans) now enable closed-loop evaluation at individual and group levels (Park et al., 2023). Grounded in coactive learning and interactive teaching, this integration aims for systems that not only perform well but also maintain alignment and shared understanding as tasks, users, and contexts evolve.

Conclusion

Intelligence in the 21st century is best understood as complementarity, a joint capability distributed across humans and machines. This perspective shifts the goal from replicating human cognition to designing systems that align with it: sharing representations where it matters, learning in ways that respect human constraints and goals, and making decisions that remain intelligible to people in real contexts. We have argued that integrating cognitive AI (mechanism-rich, interpretable, human-grounded) with machine AI (scalable, high-performance, data-driven) is the most promising route to that end.

We synthesized four concrete routes for integration and illustrated them with current initial examples of the ideas (Table 2). Together they move the field from behavioral alignment alone toward representational and process alignment: mapping machine embeddings to psychologically meaningful spaces; translating naturalistic instructions into model-usable cognitive structures; using cognitive models to supply priors, rewards, and synthetic data that shape machine learning training; and closing the loop with coadaptive agents that update policies as human states and contexts evolve (Fig. 1). Some routes are already producing measurable gains (e.g., embedding integration); others are underexplored but poised to unlock robust generalization, transparency, and sustained human-AI teamwork.

Realizing this agenda requires a practical research program. First, evaluations must be multilevel. Performance and cognitive validity should be reported, and representational geometries, learning dynamics, and decision policies should be compared with human data. Second, training signals should be cognitively shaped through priors, constraints, and synthetic corpora that encode human generalization, attention, and trade-offs. Third, systems should be built as closed loops in which cognitive models track user state and goals while machine learning components adapt perception and planning over time. Last, governance and reporting should make these mechanisms visible, enabling auditability, pedagogy, and trust.

If we treat cognitive and machine AI as mutually constraining parts of a single system, we can construct AI that not only performs at scale but also reasons with us, learns from us, and remains aligned as conditions change. The outcome is not smarter machines or more accurate cognitive models alone but a new level of human-AI complementary intelligence that advances capability while safeguarding human agency in complex sociotechnical settings.

Recommended Reading

Gonzalez, C., & Heidari, H. (2025). (See References). Distinguishes between cognitive and machine AI and provides a framework to move forward toward an integrated form of human-AI collaboration that would produce complementarity.

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Vaccaro, M., Almaatouq, A., & Malone, T. W. (2024). (See References). Presents a meta-analysis of human-AI studies to demonstrate that complementarity is generally not achieved, providing some venues for future research.

Transparency

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ORCID iDs

Cleotilde Gonzalez https://orcid.org/0000-0002-6244-2918
Tailia Malloy https://orcid.org/0000-0003-2512-984X

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T. Malloy is now at TruX, University of Luxemburg.

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